COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	8/13/26	**DATE SUBMITTED:**	
NAME:	Pizarro Jeus	**INSTRUCTOR:**	

ACTIVITY NO. ___ PRELIM EXAM

1. PROCEDURE OUTPUT

What procedures did you perform?

hostname	Operating system	Ip address
pizza1	Ubuntu	192.168.56.111
pizza2	Ubuntu	192.168.56.110
pizza3	Ubuntu	192.168.56.112

Part A: Installing and Verifying Ansible

```
pizza1@pizza1-local:~/Desktop$ ansible --version
ansible [core 2.17.14]
  config file = None
  configured module search path = ['/home/pizza1/.ansible/plugins/modules', '/usr/share/ansible/plugin
  ansible python module location = /home/pizza1/.local/lib/python3.10/site-packages/ansible
  ansible collection location = /home/pizza1/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
```

```
pizza1-local:~/Desktop$ ssh pizza3@192.168.56.112
192.168.56.112's password:
 to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

mentation:  https://help.ubuntu.com
gement:     https://landscape.canonical.com
ort:        https://ubuntu.com/pro

 Security Maintenance for Applications is not enabled.

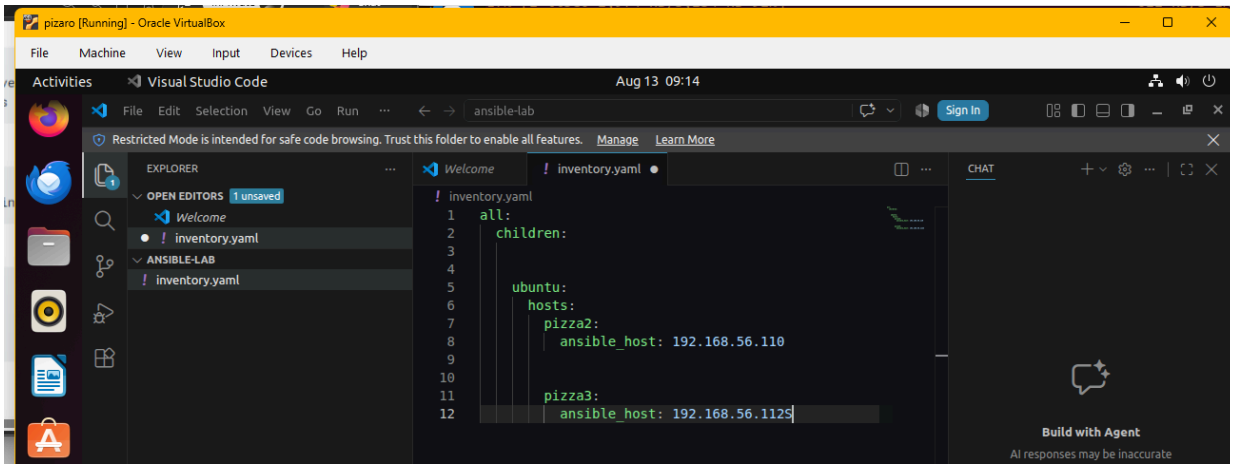
tes can be applied immediately.
hese updates are standard security updates.
these additional updates run: apt list --upgradable

ESM Apps to receive additional future security updates.
ps://ubuntu.com/esm or run: sudo pro status

ease '24.04.4 LTS' available.
-release-upgrade' to upgrade to it.

gin: Thu Aug 13 08:27:15 2026 from 192.168.56.111
```

Part B:




```
pizza1@pizza1-local:~/ansible-lab$ ansible all -i inventory.yaml --list-hosts
  hosts (2):
    pizza2
    pizza3
pizza1@pizza1-local:~/ansible-lab$ ss
```

```
pizza1@pizza1-local:~/ansible-lab$ ansible all -m ping -i inventory.yaml
[WARNING]: Platform linux on host pizza2 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
pizza2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
```

```
[WARNING]: Platform linux on host pizza3 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
pizza3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
```

PART C:

```
pizza1@pizza1-local:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yam
l
[WARNING]: Platform linux on host pizza3 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
pizza3 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host pizza2 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
pizza2 | SUCCESS => {
    "ansible_facts": {
```

```
}
pizza1@pizza1-local:~/ansible-lab$ ansible linuxservers --list-hosts -i inventor
y.yaml
  hosts (2):
    pizza2
    pizza3
pizza1@pizza1-local:~/ansible-lab$
```

Part D:

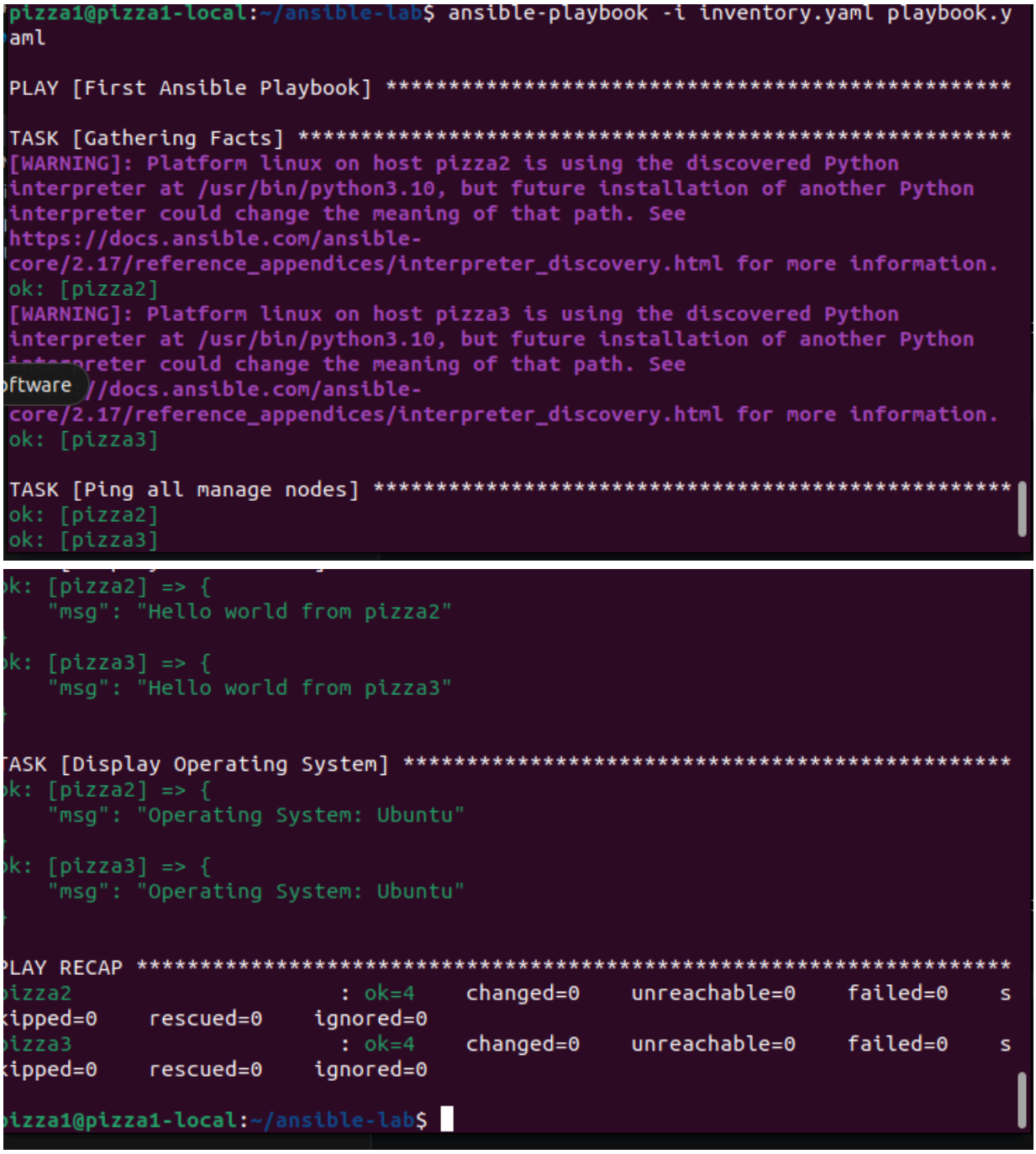
```
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [pizza2]
[WARNING]: Platform linux on host pizza3 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [pizza3]

TASK [Show Operating System] ***************************************************
ok: [pizza2] => {
    "msg": "Host pizza2 is running Ubuntu"
}
ok: [pizza3] => {
    "msg": "Host pizza3 is running Ubuntu"
}

PLAY RECAP *********************************************************************
pizza2                     : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
pizza3                     : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

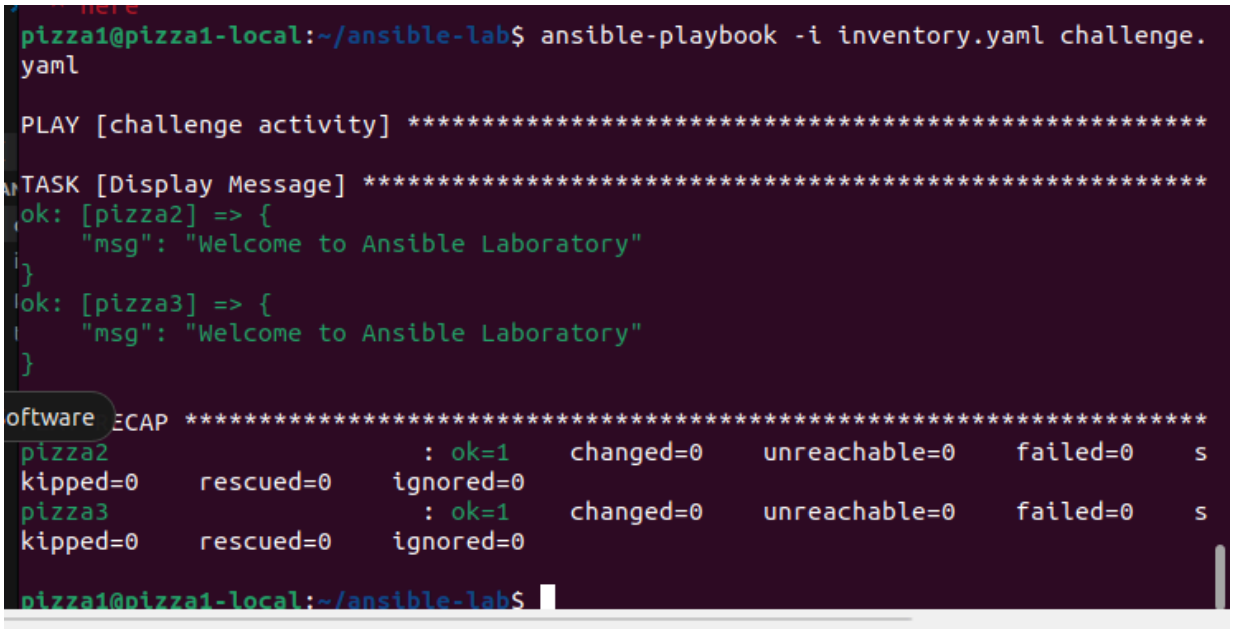
pizza1@pizza1-local:~/ansible-lab$ s
```

PART E:



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.



Guide Questions

1. What is the role of the Control Node in Ansible?
   - THE CENTRAL MACHINE WHERE ANSIBLE IS INSTALLED TO MANAGE AND PUSH TASKS TO TARGET HOSTS
2. Why is an inventory file required?
   - ITS REQUIRED BECAUSE OT ;OST CATEGORIZES AND TRACKS ALL TARGER HOSTS
3. What is the purpose of a metagroup?
   - IT AGGREGATES MULTIPLE DISTINCT HOST GROUPS INTO A SINGLE PARENT GROUP ENABLING BROAD AUTOMATION ACROSS LARGE SEGMENTS AT ONCE
4. How do variables simplify Ansible administration?
   - BY SEPARATING DYNAMIC VALUES FROM EXECUTION LOGIC, ALLOWING A SINGLE PLAYBOOK TO BE REUSED ACROSS DIFF ENVIRONMENTS
5. What is the difference between an ad hoc command and a playbook?
   - AD HOC IS QUICK, SINGLE LINE CLI TOOLS FOR ONE-OFF TASKS, WHILE PLAYBOOK ARE REUSABLE YAML SCRIPTS THAT DEFINE COMPLEX MULTI-STEP AUTOMATION

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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